

Orientation Map — creative-automation-pipeline

Glance before the call. Each: what it does / the one decision or gotcha. Source of truth is the code.

routes/campaign.ts

POST /generate, /status, /manifest, /compliance, image serving.

GOTCHA: /generate is fire-and-forget; returns runId instantly, run dies with the process.

workflow/campaignWorkflow.ts

Declares the 6 steps via .then()...commit().

GOTCHA: purely linear, no branch/parallel. Order is the design.

workflow/mastra-compatible.ts

Hand-rolled shim mimicking Mastra createStep/createWorkflow.

GOTCHA: API-shape compatible only. In-memory linear runner, no suspend/resume/persistence.

lib/runStore.ts

In-memory Map of run state + cost accumulator + runReport.

GOTCHA: ephemeral Map, self-deletes after 1hr. Crash = total loss. Weakest spot.

lib/retry.ts

Exponential backoff + jitter, regex error classification.

GOTCHA: only wraps Gemini + Dropbox. Terminal msg hardcoded "Image generation failed" even for Dropbox.

steps/loadBrief.ts

Parse + Zod-validate YAML/JSON brief.

GOTCHA: requires min 2 products; brandPalette + prohibitedWords optional.

steps/gatherAssets + agents/assetGatherer.ts

Reuse existing asset or generate via Gemini.

GOTCHA: not an LLM agent, just a for-loop. Re-throws on first product failure (no isolation).

steps/renderAspectRatios.ts

Sharp resize to 1x1 / 9x16 / 16x9, cover-fit. Writes base.png.

GOTCHA: sequential per product+ratio, no parallelism.

steps/applyOverlay.ts

SVG message bar (auto-shrink font) + optional logo. Writes final.png.

GOTCHA: the black bar it adds later pollutes the color compliance check.

steps/organizeOutputs.ts

Verify files, upload via storage adapter, write manifest.json.

GOTCHA: upload failure is soft. Run still "done", falls back to local path.

steps/checkCompliance + agents/complianceChecker.ts

Color + logo + legal-word scan.

GOTCHA: per-image try/catch, the only resilient step. Drill the three scorers.

tools/extractColors.ts

50x50 quantized dominant colors, Euclidean palette score, pass ≥ 0.3.

GOTCHA: empty palette = score 0 = every image FAILS color compliance.

tools/checkLogoPresence.ts

Claude vision yes/no on any logo/text/watermark.

GOTCHA: no reference matching, no retry, errors swallowed to false. Campaign text reads as "logo present."

tools/generateAssetWithGenAI.ts

Claude crafts prompt, Gemini renders image.

GOTCHA: image cost hardcoded \$0. The Claude prompt call is NOT retried.

lib/ai-clients.ts

Anthropic client factory + model constants.

GOTCHA: TEXT_MODEL = claude-sonnet-4-5. No temperature set anywhere (non-determinism).

lib/paths.ts

Resolves OUTPUT_DIR / BRIEFS_DIR / LOGS_DIR.

GOTCHA: OUTPUT_DIR is ONE shared dir, not per-run. Concurrent runs clobber each other.

lib/storage/

Adapter interface. Local + Dropbox live, S3/Azure throw.

GOTCHA: Dropbox chosen for creative-team workflow, not tech preference. Have that reason ready.

campaign-logger.ts / progress-bus.ts

Pino structured logs. EventEmitter feeds runStore.

GOTCHA: cost ledger has no per-call latency field. Latency is run-level only.

lib/db/

Empty Drizzle stub, never imported.

GOTCHA: this IS your "Postgres later." Honest deferral, not a lie.

frontend/src/App.tsx

Polls /status every 750ms until done/error.

GOTCHA: polling because Replit proxy buffers SSE. No SSE endpoint exists.

ONE-BREATH SUMMARY

6 linear Zod-gated steps, in-process and in-memory. Two deterministic compliance gates (color, legal words) plus one advisory vision check (logo). **Strengths:** per-step contracts, structured observability, vendor seams. **Three holes to own first:** (1) ephemeral run state = crash loses everything, (2) shared output dir = concurrency clobber, (3) thin compliance scoring.